Project Design Phase-I
Solution Architecture

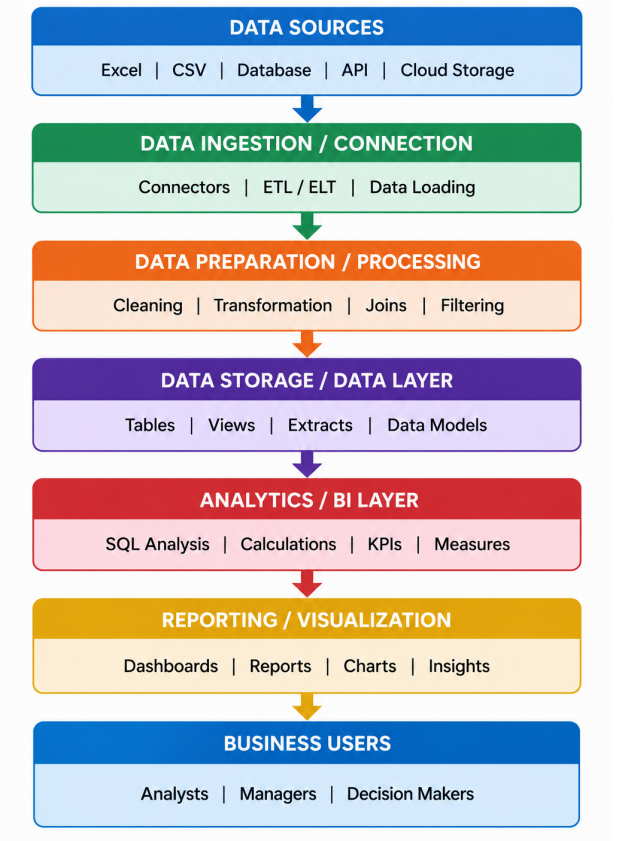
Date	28 October 2026
Team ID	SKILLWALLET-EV-01
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Sample Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technology Used

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Data Source Layer	Raw dataset containing 100,000 global electric vehicle specifications across 37 attributes.	CSV (ev_market_test.csv)
Data Ingestion & Extraction	Extracts raw CSV records into optimized in-memory Tableau format for ultra-fast query execution.	Tableau Data Engine (.hyper)
Data Preparation & Business Logic	Calculates derived fields (Cost per Km Range, Battery Degradation Severity, Total CO2 Saved) and cleans types.	Tableau Calculated Fields & Python
Analytics & Visualisation Layer	Constructs 9 core visual analytics worksheets mapping sales, battery efficiency, and financial parameters.	Tableau Desktop
Consolidated Dashboard & Story	Consolidates individual visualisations into a single interactive dashboard with global slicers and narrative Story scenes.	Tableau Dashboard & Story
Web Integration & Deployment	Embeds the published interactive Tableau Public dashboard into a web portal accessible to all stakeholders.	Python Flask, HTML5, CSS3, Tableau Public